Automatic Speech Recognition for Voice Log Analysis: A Case Study in the Indonesia National Single Window Contact Center

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Abstract*— Audio interactions at contact center generate extensive voice logs containing valuable information about customer needs, recurring issues, and service performance, these data as raw audio files, remain underutilized. The Directorate of Services Data and Partnerships Management of the INSW Agency recorded 13,765 service requests in March 2025, including 3,652 incoming calls, of which 2,734 (74.9%) were successfully handled by agents. This study develops an end-to-end Automatic Speech Recognition (ASR) and Natural Language Processing (NLP) system powered by Large Language Models (LLMs) to transform INSW voice logs into actionable insights. The system integrates Faster-Whisper for transcription and GPT-4.1 Mini for topic classification, entity extraction, and summarization, orchestrated via Apache Airflow and visualized in a Streamlit dashboard. Quantitative evaluation achieved a Word Error Rate (WER) of 12.6% and Character Error Rate (CER) of 6.4% on Indonesian conversational speech, outperforming Whisper-Base (CPU) by 9%. NLP performance reached macro-F1 = 0.87 across five service topics, while summarization quality measured by ROUGE-1 = 0.79 and BERTScore = 0.83 demonstrated strong alignment with human references. These results show that leveraging LLM-based ASR pipelines can significantly enhance contact center analytics and operational efficiency while reducing manual monitoring. The proposed cost-efficient architecture highlights the feasibility of applying multilingual LLMs for public-sector call analytics, contributing to service transparency and data-driven decision-making at INSW.*Keywords— Natural Language Processing, Large Language Model, Topic Modeling. (**keywords**)**

I. INTRODUCTION

A. Introduction

The Indonesia National Single Window (INSW) Agency is a non-echelon organizational unit under the Ministry of Finance of the Republic of Indonesia. One of its key divisions, the Directorate of Services Data, and Partnerships Management, is responsible for managing regulatory information as the primary reference for customs documentation in export and import activities. It also oversees the governance of electronic data and information related to national logistics and trade, and facilitates partnerships for the development and implementation of the Indonesia National Single Window (INSW) system at both national and international levels [1].

As part of these responsibilities, the Directorate also operates the INSW Contact Center, which is responsible for providing public services by offering information related to export-import activities and/or the INSW System [2]. In March 2025, the Contact Center received 13,765 service requests, including 3,652 incoming calls, of which 2,734 (74.9%) were successfully handled by service agents [3]. These figures highlight the crucial role of the Contact Center in supporting trade facilitation while also revealing gaps in service efficiency.

Despite its importance, several challenges remain in managing the conversation data (voice logs), which reduce the effectiveness of service evaluation:

1) Voice logs stored only as audio — Conversation data are currently maintained solely in audio format without transcription, preventing further text-based analysis.
2) Lack of automated analysis tools — There is no system available to identify main topics, categorize complaints, or analyze user sentiment within the conversations.
3) Absence of integration with evaluation systems — User feedback captured in voice logs has not been incorporated into existing evaluation frameworks or monitoring dashboards, limiting its impact on service improvement.
4) Technological adoption gap — While studies show that less than 20% of contact center agents in North America remain unexposed to AI tools [4], INSW's operations still rely on manual review, indicating a gap in leveraging automation for efficiency.

Consequently, voice log data remain underutilized despite their potential to improve service evaluation. Building on global LLM-based contact center innovations [5], this study applies ASR and NLP to automatically identify call drivers and topics, enhancing analytical accuracy and efficiency. In line with SDG 9, it proposes a cost-efficient AI framework to transform Indonesian-language voice logs into actionable insights for data-driven decision-making at INSW.

B. Research Objectives

The objectives of this study are:

1) Develop an automated system to transcribe INSW voice logs using Automatic Speech Recognition (ASR) based on the Faster-Whisper model.
2) Apply Natural Language Processing (NLP) and Large Language Models (LLMs) to summarize conversations, extract key entities, and identify topics relevant to national and international trade services.
3) Integrate analysis results into a unified workflow and dashboard for real-time monitoring and decision support.
4) Evaluate system performance using ASR (WER/CER), LLM (F1, precision, recall), and summarization (ROUGE/BERTScore) metrics to assess accuracy and efficiency.

II. LITERATURE REVIEW

Recent studies highlight the growing role of Automatic Speech Recognition (ASR), Natural Language Processing (NLP), and Large Language Models (LLMs) in analyzing unstructured voice data. Arisaputra [6] improved Indonesian ASR accuracy using XLSR-53, while Tamrin et al. [7] applied ASR for real-time sentiment analysis. Lyu et al. [8] and Gokcimen et al. [9] evaluated Whisper for multilingual and noise-robust transcription, and Wang et al. [10] further examined its contextual learning capabilities.

In contact-center analytics, Embar et al. [5] proposed a cost-efficient LLM-based insight-extraction system that generates concise call drivers for topic modeling and trend detection, achieving competitive results with GPT-3.5 while reducing cost and latency. Complementary NLP research—such as Kaliuta [11] on customer-request classification and Nabiilah et al. [12] on multilabel text classification using IndoBERT/MBERT—strengthens automated text understanding. Recent LLM developments by An et al. [13] (FILM-7B) and Zhou et al. [14] in telecommunications highlight broader applications for classification and prediction.

This study extends prior work by integrating Faster-Whisper and GPT-4.1 Mini into a unified pipeline, evaluated using WER, CER, F1-score, and ROUGE/BERT Score to benchmark accuracy and efficiency in Indonesian-language contact-center analytics.

III. RESEARCH METODOLOGY

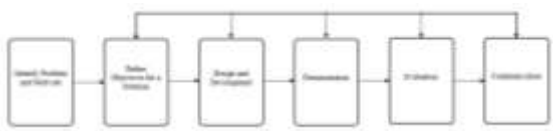

Fig. 1. Research methodology diagram

This study applies the Design Science Research Methodology (DSR) proposed by Wildenauer [15], which provides a structured framework for developing and evaluating technology-based artifacts. The process consists of six iterative stages (Fig. 1).

1. Identify Problem and Motivate

Conversation data (voice logs) at the INSW Contact Center are stored solely as audio, without transcription or analysis, limiting their value for performance evaluation and service improvement.

2. Define Objectives for a Solution

The research aims to design a system that automatically converts voice logs into text, classifies topics, extracts key entities, and summarizes conversations to support service assessment.

3. Design and Development

The system integrates Faster-Whisper for Automatic Speech Recognition (ASR) and GPT-4.1 Mini for Natural Language Processing (NLP) and Large Language Model (LLM) tasks. Its architecture includes modules for transcription, topic modeling, summarization, and entity recognition, with workflows orchestrated via Apache Airflow.

4. Demonstration

The prototype is implemented and tested on real INSW Contact Center voice logs. Results are visualized through a Streamlit dashboard, displaying call classifications, summaries, and complaint trends to support decision-making.

5. Evaluation

System performance was evaluated using ASR metrics—Word Error Rate (WER) and Character Error Rate (CER)—and NLP/LLM metrics including precision, recall, F1-score, and ROUGE/BERTScore for summarization. Latency and throughput were also measured for runtime efficiency. The WER is calculated as:

$$\text{WER} = \frac{S + D + I}{N}$$

where S,D, and denote substitutions, deletions, and insertions, and N is the total number of reference words. A lower WER indicates higher transcription accuracy.

6. Communication

Research results are disseminated through this thesis report, academic defense, and internal recommendations to INSW stakeholders, ensuring the practical application of AI-driven analytics within the agency's operational framework.

IV. RESULT AND DISCUSSION

A. *System Architecture Design*

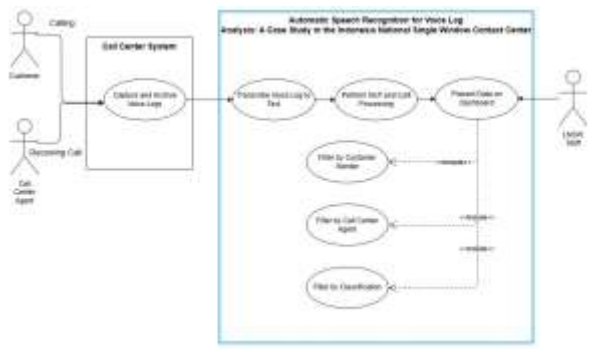

Fig. 2. Use case diagram of the proposed system

No	Process	Actor	Output
1	Customer contacts the Call Center	Customer	Voice conversation initiated
2	Call Center Agent receives the call	Call Center Agent	Voice conversation handled
3	LNSW Staff accesses the dashboard	LNSW Staff	Conversation data displayed
4	LNSW Staff filters by Call Center Agent	LNSW Staff	Dashboard view filtered by specific call center agent
5	LNSW Staff filters by Customer Gender	LNSW Staff	Dashboard view filtered by customer gender
6	LNSW Staff filters by Service Classification	LNSW Staff	Dashboard view filtered by service category

Fig. 3. Use case process table

The interaction between users and the system is illustrated in the use case diagram shown in Fig. 2. The system is designed to support the INSW Contact Center in analyzing voice log data using Automatic Speech Recognition (ASR), Natural Language Processing (NLP), and Large Language Models (LLMs).

The system involves three primary actors:

1) Customer – Initiates the call and provides the voice input.
2) Call Center Agent – Engages in conversation with the customer; their identity is captured and used for filtering and analysis.
3) INSW Agency Staff – Uses the dashboard to monitor, analyze, and make decisions based on the processed voice log data.

The use case diagram includes system functionalities such as capturing and archiving voice logs, transcribing audio to text, performing NLP and LLM-based analysis, and presenting the results through an interactive dashboard. The dashboard supports filtering by agent, customer gender, and service classification.

The detailed interaction steps between actors and the system are outlined in Fig. 3., which presents the use case process. Each step describes the actor's role and the expected system output. For instance, when the INSW Agency staff filters data by classification, the dashboard dynamically updates to display only the relevant conversations.

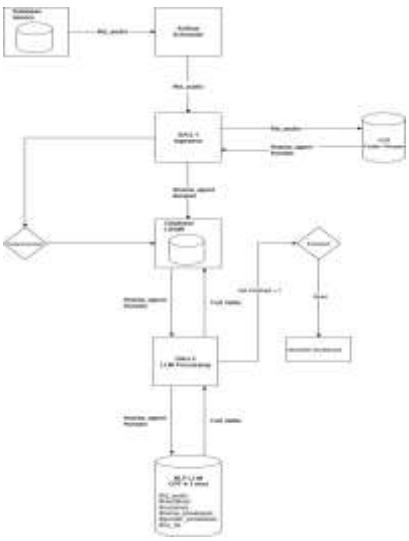

Fig. 4. Flowchart voice log analytic pipeline of the proposed system

This research designs an integrated information system to automatically analyze voice log the INSW Contact Center service. The system consists of several main stages as illustrated in Fig. 4. :

1) Audio Retrieval and Storage

Audio Retrieval and Storage Voice logs (.wav/.mp3) are automatically retrieved from the vendor's database and stored in a staging table within INSW's PostgreSQL. In this research, 41 audio logs with an average duration of 5–7 minutes (recorded in 2022) were used as the primary dataset for system testing and validation.

2) Automatic Speech Recognition (ASR)

Audio files are processed using the Faster-Whisper model to convert speech into text. The transcription results, including the agent's name, are stored in the main table.

Fig. 5. Bahasa audio transfer to text

3) NLP and LLM Analysis

The transcribed text is analyzed using *GPT-4.1* Mini to classify topics, generate summaries, and extract key entities such as caller name, gender, and reference number.

Caller: Halo, saya mau tanya soal dokumen SSM saya yang duakali kena reject.
Agent: Baik, Bapak bisa sebutkan nomor ajunya?
Caller: Nomornya <nomor_aju>.
Agent: Terima kasih, saya bantu cek dulu ya. Mohon ditunggu sebentar.
Caller: Iya, Bu.
Agent: Dari hasil pengecekan, dokumen wajib sudah ada semua, tapi ada dokumen lain seperti surat tujuan pemastian dan COO.
Caller: Oh iya, saya sudah upload semuanya. Tapi kok masih reject ya?
Agent: Betul, tampaknya ada kendala sinkronisasi antara sistem INSW dan sistem karantina.
Caller: Jadi harus bagaimana, Bu?
Agent: Bapak bisa kirim ulang dokumen, lalu hubungi pihak karantina agar dicek kembali.
Caller: Baik, saya coba kirim ulang. Terima kasih, Bu.
Agent: Sama-sama, Bapak. Semoga prosesnya lancar.

TABEL I. SUMMARY, TOPIC CALSSIFICATION AND ENTITIES

Topic	Summary	Caller Name	Caller Gender
SSM Izin	The caller reported that his SSM transfer was rejected twice because the 4.2 COO document was not uploaded correctly in the quarantine system. The agent confirmed the required documents were complete but found a synchronization issue between INSW and the quarantine system, advising the caller to re-upload the document and follow up if the problem continued.	Mr. A***	Male

4) Process Orchestration with *Apache Airflow*

Process Orchestration – All workflows are managed through Apache Airflow, ensuring consistent scheduling, error recovery, and prevention of duplicate processing via status tagging (finished=0/1).

5) Data Visualization

The analysis results are displayed in an interactive dashboard built with *Streamlit*, facilitating service trend monitoring and supporting data-driven decision-making.

B. *Implementation of Voice Log Analytics Pipeline*

TABEL II. VOICE LOG IMPLEMENTATION STEPS

Activity	Activity Description
Initial ASR Algorithm Implementation and Experimentation	Built the initial Faster-Whisper integration in Python, configured the local ASR environment, and installed required dependencies.
Source Code Understanding and Algorithm Exploration	Analyzed Whisper structure, inputs, and parameters; tested parameter variations including model size, language, beam size, and floating-point precision (fp16/fp32).
Audio Extraction and ASR Pipeline	Extracted audio files for ASR testing; compared transcription formats and model performance in terms of speed and accuracy.

1) Ingestion and Preprocessing

The first stage of the system involves automatic data ingestion from the INSW Call Center vendor database, as outlined in Table I. Using Apache Airflow, the ingestion task is scheduled daily at 00:00 Western Indonesian Time (UTC+7). Airflow logs each process execution and transfers retrieved voice logs to a staging table in the INSW PostgreSQL database for processing. Preprocessing is then applied to standardize and enhance audio quality prior to transcription. Key steps include:

- Metadata standardization, such as agent ID, date, and timestamp normalization.Cleaning noise using audio libraries to improve sound quality.
- Segmenting audio (if needed) to separate sessions or speakers.

A total of 41 anonymized voice logs from 2022 were used for experimentation, with an average duration of 5–7 minutes per file. These preprocessing steps ensured consistent input quality and metadata uniformity before the ASR stage.

2) *Automatic Speech Recognition* (ASR) Pocess

```
from faster_whisper import WhisperModel
# Initialize the model
model_size = "small"
model = WhisperModel(model_size, device="cpu",
compute_type="int8")

# Transcribe the audio file
    segments, info = model.transcribe(full_path, language="id")

# Combine segments into full transcription
    uraian = ""
    for segment in segments:
        uraian += f"{segment.text.strip()}\n"

    savetosqlite("database_log.db", [(nama_agent, uraian, id_audio)])
    print(f"Transcription for {f} completed.")
    move_to_processed(f)
```

Fig. 6. Example script for automatic speech secognition (ASR)

ASR represents the initial stage of transforming audio into text. This process is automated and orchestrated by Apache Airflow to ensure reproducibility and logging consistency. The Faster-Whisper model [13] was selected for its balance of high accuracy in Indonesian speech and computational efficiency for CPU-based local inference.

The operational flow is as follows:

- Audio Retrieval: Airflow fetches unprocessed audio files from the PostgreSQL staging table.
- Audio Transcription: Faster-Whisper converts audio into Indonesian text while preserving conversational context.
- Initial Metadata Entry: The text is inserted into the description column, and the agent's name into the name_agent column. Other columns remain empty.
- Status Update: Data is saved with finished = 0, indicating ASR is complete but NLP analysis has not yet been performed.

3) Natural Language Processing (NLP) and Large Language Model (LLM) Utilization

```
prompt_summary = """Extract elemen informasi dari percakapan berikut ini:
nama agent, nama penelepon, gender penelepon, summary percakapan, nomor
car / nomor aju (26 digit), klasifikasi (eCOO, SSM Izin, SSM QC, PIB, PEB,
dan N/A jika tidak ada)

Return dalam format JSON seperti:
{
 "data": {
   "summary": "...",
   "tanggal": "...",
   "nama_penelepon": "...",
   "gender_penelepon": "...",
   "no_aju": "...",
   "klasifikasi": "..."
 }
}
Percakapan:
"""
```

Fig. 7. Prompt GPT 4.1 for NLP and LLM

Following ASR, transcribed text is processed through NLP and LLM modules to generate structured analytical insights. The GPT-4.1 Mini model is used to classify topics, summarize conversations, and extract named entities. The key section of the script used for this process is shown in Fig. 7.

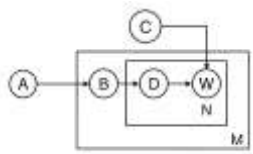

Fig. 8. Latent Dirichlet Allocation (LDA) generative model

NLP and LLM process stages:

- Retrieving Incomplete Data: Airflow retrieves records from the hasil_log table lacking summaries or analytical metadata
- Conversation Topic Classification**:** As shown in Fig. 8, the adapted LDA model represents each conversation as a mixture of topics B generated from a Dirichlet prior A, with word distributions C [16]. For each word W, a latent topic D is sampled across N words and M conversations. The resulting clusters like ekspor, impor, e-

COO, registrasi, and SSM QC are refined using few-shot GPT-4.1 Mini, achieving consistent labeling with IAA = 0.91.

- Named Entity Recognition (NER): The system extracts entities such as institution names, document numbers, caller names, genders, and reference numbers (*no_aju*).
- Conversation Summarization: Prompt-based summarization generates concise explanations of each caller's intent.
- Text Cleaning and Normalization: Linguistic noise is removed, spelling is corrected, and formatting is standardized to align with INSW NLP conventions.
- Storing Analysis Results: All analytical outputs topic classifications, entities, and summaries are saved to the *hasil_log* table, with the record status updated to finished = 1, indicating completion of the NLP and LLM processes.

C. *Integration of Daily Data Flow Workflow*

The INSW voice log processing system is designed to be automated and scheduled using *Apache Airflow*, which manages two main workflows: (*Directed Acyclic Graph*) *DAG-1 Ingestion and (Directed Acyclic Graph) DAG-2 LLM Processing.*

Every day at 00:00 Western Indonesian Time (WIB, UTC+7), *DAG-1* pulls audio data from the vendor database and stores it in the *PostgreSQL* staging table along with initial metadata. The audio is then processed by the *Faster-Whisper* ASR model to generate transcriptions and agent names, and saved with finished = 0.

DAG-2 processes incomplete data using *GPT-4.1* Mini to classify topics, extract entities (such as caller name and *no_aju*), and generate conversation summaries. Once all columns are filled, the status is updated to finished = 1, indicating the data is ready for analysis in the *Streamlit dashboard.*

D. *Data Storage and Visualization*

Transcription and analysis results are stored in the INSW Agency *PostgreSQL* database, specifically in the *hasil_log* table as the main repository. Each entry has a unique ID and contains the transcript, topic classification, summary, key entities, and metadata such as date and call duration. A status of finished = 1 indicates the data is ready for analysis, while finished = 0 will be automatically reprocessed.

Fig. 9. Example of visualization using Streamlit: classifications

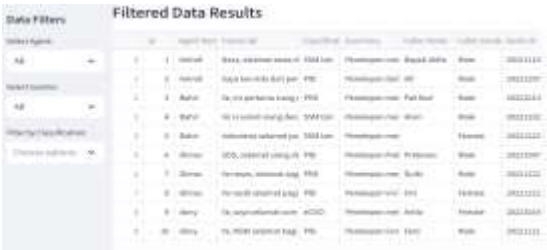

Fig. 10. Example of visualization using Streamlit: filter data

The visualization is implemented through a Streamlit-based dashboard designed to be user-friendly. Its main features include:

1) Call Statistics by Service Category
   Displays the distribution of conversation topics based on INSW business process classifications, helping to understand users' primary concerns. This feature is illustrated in Fig. 7.

2) Conversation Trends Over Time
   Visualizes changes in the pattern of questions or complaints, useful for detecting shifts in user needs.

3) Call Driver Summary
   Provides a list of call reasons that can be filtered by time period or specific agents, supporting operational analysis. This feature is illustrated in Fig. 8.

4) Entity Analysis
   Highlights key entities such as company names or important submission numbers, aiding in the identification of recurring patterns or specific issues.

5) Filtering and Search Features
   Allows users to filter data by date range, service category, or specific keywords, enhancing flexibility in data exploration.

E. *Discussion*

This study developed an automated system to analyze INSW Contact Center voice logs, transforming raw audio into structured insights. Using the Design Science Research (DSR) approach, the system integrates Faster-Whisper ASR, NLP, and LLMs within an Apache Airflow workflow and Streamlit dashboard to automate transcription, topic classification, and entity extraction. Results show that ASR–LLM integration can effectively support Indonesian-language voice-log analysis and enhance data-driven service evaluation.

Several limitations remain. The accuracy of transcription and classification still requires formal benchmarking, and the system lacks noise reduction, dialect handling, and voice masking features. CPU-only processing also limits scalability for long calls and high daily volumes, indicating the need for runtime optimization and GPU support.

Regarding privacy and security, all 2022 voice logs were anonymized, API keys managed via encrypted environment variables, and workflows logged in Apache Airflow for traceability. Future work will focus on dataset expansion, real-time processing, and stronger data protection and access controls, improving scalability, privacy compliance, and interoperability for public-sector analytics.

V. CONCLUSION

This study developed an automated system for analyzing INSW Contact Center voice logs, transforming previously unprocessed audio into structured insights. Using the Design Science Research (DSR) approach, the system integrates Faster-Whisper ASR, NLP, and LLMs within an Apache Airflow workflow and Streamlit dashboard for transcription, topic classification, and trend visualization.

The results confirm that ASR–LLM integration effectively automates Indonesian-language voice-log analysis, supporting data-driven service evaluation. Future work includes real-time deployment, larger and more diverse datasets, and extended features such as sentiment analysis, business-intelligence integration, and enhanced data security to strengthen scalability and inter-agency collaboration.

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